Frame It as Cases: Work That Speaks for Itself

Voice Card:

Direct, honest, practical, clear, professional, no buzzwords.

Case Study 1

<u>Secure RAG</u>

Project:

Secure RAG — Enterprise AI Document Intelligence System

Problem:

Organizations often store important information across multiple documents, making it difficult for employees to quickly find accurate answers. I wanted to build an AI system that could retrieve information efficiently while maintaining enterprise-level security.

What I Did:

I designed and developed SecureRAG using Retrieval-Augmented Generation (RAG), ChromaDB, Sentence Transformers, and the Claude API. I implemented semantic document retrieval, prompt injection detection, toxic content filtering, query validation, and audit logging. While building the project, I chose a rule-based security layer because it was transparent, easy to explain, and suitable for demonstrating secure AI system design.

Outcome:

The project demonstrated how enterprise AI document assistants can combine accurate information retrieval with security guardrails. Although it is a portfolio prototype, it strengthened my understanding of secure AI system architecture, vector databases, enterprise AI workflows, and responsible AI development.

Case Study 2

<u>MCP Powered AI Assistant</u>

Project:

MCP-Powered AI Assistant

Problem:

Modern AI assistants often depend on manually configured tools. I wanted to explore how an AI system could automatically discover and select the appropriate tool for different user requests.

What I Did:

I developed an AI assistant using Anthropic's Model Context Protocol (MCP), Llama 3.3 70B through the Groq API, and multiple production-ready tools including web search, calculator, and file management. I also implemented conversation memory and tool activity logging to improve usability and transparency.

Outcome:

The project demonstrated dynamic tool orchestration, context-aware reasoning, and modular AI assistant design. It improved my understanding of AI agents, MCP architecture, and multi-tool workflows.

Case Study 3

<u>Sentinel AI</u>

Project:

Sentinel AI — Deepfake Detection Platform

Problem:

Deepfake technology is becoming increasingly sophisticated, creating risks for digital identity verification and online trust. I wanted to build a system capable of detecting manipulated visual content.

What I Did:

I designed a multimodal deepfake detection platform combining Vision Transformers, CNNs, OpenCV, and Grad-CAM explainability. I researched transformer architectures, evaluated model performance, and used Grad-CAM to understand how the model reached its predictions.

Outcome:

The project enhanced my understanding of computer vision, explainable AI, transformer models, and AI model evaluation while demonstrating practical approaches for AI-assisted fraud detection.

About Me:

I am a final-year B.Tech Artificial Intelligence and Data Science student passionate about building practical AI applications using Machine Learning, Large Language Models, Retrieval-Augmented Generation (RAG), and modern AI engineering practices. Through internships and personal projects, I enjoy solving meaningful problems, learning emerging AI technologies, and continuously improving my engineering skills by building practical AI applications that address real-world problems.

Contact:

Interested in discussing AI projects, internships, or collaboration opportunities?

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Before vs After:

Before:

I am a results-driven AI developer passionate about innovation and cutting-edge technology.

After:

I build practical AI applications by combining Machine Learning, Large Language Models, and modern AI engineering techniques. My portfolio explains the problems I worked on, the decisions I made, and what I learned from building each solution.

AI Reflection:

To create these case studies, I used Claude as a thinking partner rather than a content writer. Claude interviewed me one question at a time about each project, challenging me to explain the problem, justify my technical decisions, and describe the outcomes honestly. This process helped me remove generic descriptions, improve clarity, and ensure every case study reflects my own experience and voice.